

Eros Fanì

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Professional and teaching experience

Machine Learning Researcher

Oct 2025 – Jul 2026

GENSYN

- Led research on **model merging**, **mixture of experts**, **LLMs routing**, and **decentralized ML**
- Designed and implemented a **decentralized AI inference platform** that orchestrates multiple remote LLM providers behind a unified conversational interface (public release forthcoming)

Postdoctoral Fellow & Software Engineer

Nov 2024 – Oct 2025

BASQUE CENTER FOR APPLIED MATHEMATICS (BCAM)

- Lead developer, **Act.AI project** (PI: Prof. N. Quadrianto)

Senior researcher

May 2024 – Oct 2025

PREDICTIVE ANALYTICS LAB, UNIVERSITY OF SUSSEX (UoS)

- Led research on mitigating the effects of **spurious correlations** in **federated learning**, and on the development of a **unified framework** for LLMs for **tabular data**

MSc thesis supervisor

Sep 2023 - Oct 2024

POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

- Topic: Clustered Federated Learning
- Resulted in a publication at **ICML25**

Teaching Assistant

Sep 2022 - Sep 2023

POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

- Graduate courses: Machine Learning and Deep Learning ◦ Advanced Machine Learning ◦ Data Analytics and Artificial Intelligence ◦ Web Applications I
- Undergraduate course: Computer Science

MSc thesis supervisor

Sep 2022 - Jul 2023

POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

- Topic: Federated Visual Geo-Localization
- Resulted in a publication at **FedVision @ CVPR 2024**

Teaching collaborator as an undergrad

Sep 2018 - Mar 2019

POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

- Undergraduate courses: Algorithms and Programming ◦ Mathematical Analysis I ◦ Mathematical Analysis II

Education

Ph.D. in Computer and Control Engineering

Nov 2021 - Jul 2025

VISUAL AND MULTIMODAL APPLIED LEARNING LABORATORY (VANDAL), POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

- Thesis title: “**Addressing Heterogeneity in Federated Learning for Real-world Vision Applications**”
- Advisor: Prof. B. Caputo · Polytechnic University of Turin (PoliTo)

ELLIS Ph.D. program - Academic track

Nov 2021 - Jul 2025

EUROPEAN LABORATORY FOR LEARNING AND INTELLIGENT SYSTEMS (ELLIS)

- Primary Host: Prof. B. Caputo · Polytechnic University of Turin (PoliTo)

- Exchange Host: Prof. N. Quadrianto · University of Sussex (UoS)

Visiting period at the University of Sussex for the ELLIS Exchange

May 2024 - Oct 2024

UNIVERSITY OF SUSSEX (UoS), EUROPEAN LABORATORY FOR LEARNING AND INTELLIGENT SYSTEMS (ELLIS)

- Research Advisor: Prof. N. Quadrianto · University of Sussex (UoS)

Visiting period at ELLIS Alicante

Sep 2024

ELLIS ALICANTE, EUROPEAN LABORATORY FOR LEARNING AND INTELLIGENT SYSTEMS (ELLIS)

- Research Advisor: Prof. N. Oliver · ELLIS Alicante

MSc in Data Science and Engineering

Oct 2019 - Oct 2021

POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

- Final grade: **110/110 cum laude**
- Thesis title: **On the Challenges of Class Imbalance in Federated Learning for Semantic Segmentation**

BSc in Computer Engineering

Oct 2016 - Jul 2019

POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

- Final grade: **108/110**

Young Talent Project

Oct 2016 - Jul 2019

POLYTECHNIC UNIVERSITY OF TURIN (POLITO), CRT FOUNDATION

- Study program of excellence open only to the 2% top-tier students admitted to the PoliTo for the A.Y. 2016/17

High school diploma

Sep 2011 - Jul 2016

LICEO SCIENTIFICO STATALE M. CURIE GIULIANOVA

- Final grade: **100/100 cum laude**

Additional courses

Warm up your muscles: a ten-stage marathon to become the protagonist of your financial future [🔗](#)

Feb 2025 - Apr 2025

MUSEO DEL RISPARMIO, INTESA SAN PAOLO

Fundamentals of Reinforcement Learning [🔗](#)

Feb 2022

COURSERA, UNIVERSITY OF ALBERTA

Sample-based Learning Methods [🔗](#)

Jan 2022

COURSERA, UNIVERSITY OF ALBERTA

Security Soft Start [🔗](#)

Jun 2020

ONSTAIRS ACADEMY

Attended schools and programs

Mediterranean Machine Learning (M2L) Summer School

Thessaloniki, Greece
Aug 2023

AI EDUCATION FOUNDATION

ELLIS Doctoral Symposium 2021

Tübingen, Germany
Oct 2021

UNIVERSITY OF TÜBINGEN, MAX-PLANCK INSTITUTE FOR INTELLIGENT SYSTEMS

Vodafone IoT Academy

Turin, Italy
Jan 2020 - Jul 2020

VODAFONE, POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

Summer Orientation School Sant'Anna

Volterra, Italy
Jun 2015

SCUOLA SUPERIORE SANT'ANNA

Summer School for Experimental Sciences

Assergi, Italy
Jun 2014

GRAN SASSO NATIONAL LABORATORIES (LNGS) OF THE ITALIAN NATIONAL INSTITUTE FOR NUCLEAR PHYSICS (INFN)

Publications

E. Fani , O. Ersoy. “ <i>IR3DE: A Linear Router for Large Language Models</i> ”. Resource-Adaptive Foundation Model Inference (AdaptFM) workshop of the International Conference on Machine Learning, 2026.	AdaptFM-ICML26
E. Fani , O. Ersoy. “ <i>Training-Free Dynamic Upcycling of Expert Language Models</i> ”. Scaling Post-training for LLMs (SPOT) workshop of the International Conference on Learning Representations, 2026.	SPOT-ICLR26
• Extended version under review at a major ML conference.	
G. D. Németh, E. Fani , et. al. “ <i>Diversity-Driven Learning: Tackling Spurious Correlations and Data Heterogeneity in Federated Models</i> ”. Third IEEE International Conference on Federated Learning Technologies and Applications, 2025.	FLTA25
E. Fani , et. al. “ <i>Resource-Efficient Personalization in Federated Learning With Closed-Form Classifiers</i> ”. IEEE Access, 2025	IEEE ACCESS
A. Licciardi, ..., E. Fani , et. al. “ <i>Interaction-Aware Gaussian Weighting for Clustered Federated Learning</i> ”. Forty-second International Conference on Machine Learning, 2025.	ICML25
E. Fani , et. al. “ <i>Accelerating Heterogeneous Federated Learning with Closed-form Classifiers</i> ”. Forty-first International Conference on Machine Learning, 2024.	ICML24
M. Dutto, ..., E. Fani , et. al. “ <i>Collaborative Visual Place Recognition through Federated Learning</i> ”. FedVision Workshop in Conjunction with the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024.	FEDVISION CVPR24
E. Fani , et. al. “ <i>Fed3R: Recursive Ridge Regression for Federated Learning with strong pre-trained models</i> ”. International Workshop on Federated Learning in the Age of Foundation Models in Conjunction with NeurIPS, 2023.	FL@FM NEURIPS23
E. Fani , et. al. “ <i>FedDrive v2: an Analysis of the Impact of Label Skewness in Federated Semantic Segmentation for Autonomous Driving</i> ”. 5th Italian Conference on Robotics and Intelligent Machines, 2023.	I-RIM23
D. Shenaj*, E. Fani *, et. al. “ <i>Learning Across Domains and Devices: Style-Driven Source-Free Domain Adaptation in Clustered Federated Learning</i> ”. IEEE/CVF winter conference on applications of computer vision, 2023.	WACV23
L. Fantauzzo*, E. Fani *, et. al. “ <i>FedDrive: Generalizing Federated Learning to Semantic Segmentation in Autonomous Driving</i> ”. IEEE/RSJ International Conference on Intelligent Robots and Systems, 2022.	IROS22

*Equal contribution.

Peer-reviewing

Conference on Neural Information Processing Systems (NeurIPS)	2026, 2025, 2024
Resource-Adaptive Foundation Model Inference workshop of the International Conference on Machine Learning (AdaptFM-ICML)	2026
Scaling Post-training for LLMs workshop of the International Conference on Learning Representations (SPOT-ICLR)	2026
IEEE Robotics and Automation Letters (RA-L)	2025, 2022
European Conference on Artificial Intelligence (ECAI)	2025
• Invited as Program Committee Member	
IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR)	2025
Transactions on Machine Learning Research (TMLR)	2024
European Conference on Computer Vision (ECCV)	2024
• Awarded as Outstanding Reviewer	
IEEE International Conference on Development and Learning (ICDL)	2023
European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)	2023

IEEE International Conference on Robotics and Automation (ICRA)	2023
Learning from Limited and Imperfect Data workshop of the European Conference on Computer Vision (L2ID-ECCV)	2022

Invited talks

Speaker in a Gensyn X Space ↗ Presenting the research paper: IR3DE: A LINEAR ROUTER FOR LARGE LANGUAGE MODELS Engaging in live Q&A with audience	Jun 2026
Speaker in a Gensyn X Space ↗ Presenting the research paper: TRAINING-FREE DYNAMIC UPCYCLING OF EXPERT LANGUAGE MODELS Engaging in live Q&A with audience	May 2026
Video presentation at the Flower AI summit, organized by Flower AI ↗ Presenting the research paper: TRAINING-FREE DYNAMIC UPCYCLING OF EXPERT LANGUAGE MODELS	Apr 2026

Honors, awards and achievements

Outstanding Reviewer award at ECCV24 EUROPEAN CONFERENCE ON COMPUTER VISION 2024, EUROPEAN COMPUTER VISION ASSOCIATION (ECVA)	Sep 2024
Scholarship award for the participation to the Mediterranean Machine Learning (M2L) Summer School AI EDUCATION FOUNDATION Selected as one of the top 10 students among 805 applicants (top 1.25%) and 180 participants (top 5.55%)	Aug 2023
Granted the WACV23 registration fee waiver IEEE/CVF WINTER CONFERENCE ON APPLICATIONS OF COMPUTER VISION 2023	Nov 2022
Two-times RoboMensa National Winner MENSA ITALIA – THE HIGH I.Q. SOCIETY Logic Contest - 1st place in both Individual and Teams categories	2015 - 2016
Two-times National Finalist at the Italian Mathematics Olympic Games in Teams ITALIAN MATHEMATICAL UNION	2015 - 2016
Three podiums at the RomeCup Rescue Secondary National Finals DIGITAL WORLD FOUNDATION 1st place (2016) – 2nd place (2014, 2015)	2014 - 2016
National Finalist at the Italian Informatics Olympic Games MINISTRY OF INSTRUCTION, UNIVERSITY AND RESEARCH (MIUR)	2015
National Finalist at the Mathematical Games of the Mediterranean ITALIAN ACADEMY FOR THE PROMOTION OF MATHEMATICS	2013

Technical blog posts

Look Beyond One-Size-Fits-All LLMs with IR3DE ↗ Blog post on the paper IR3DE: A LINEAR ROUTER FOR LARGE LANGUAGE MODELS	Jun 2026
Stop Multitask Training. Just DUME. ↗ Blog post on the paper TRAINING-FREE DYNAMIC UPCYCLING OF EXPERT LANGUAGE MODELS	Apr 2026
Accelerating Heterogeneous Federated Learning with Closed-form Classifiers ↗ Blog post on the paper ACCELERATING HETEROGENEOUS FEDERATED LEARNING WITH CLOSED-FORM CLASSIFIERS	May 2024

FedDrive: a Federated Learning Benchmark for Semantic Segmentation in Autonomous Driving [↗](#)

Sep 2022

Blog post on the paper FEDDRIVE: GENERALIZING FEDERATED LEARNING TO SEMANTIC SEGMENTATION IN AUTONOMOUS DRIVING

Selected media mentions ---

The Missing Dispatcher for Expert Models [↗](#)

Jun 2026

X article from Gensyn — Coverage of invited X Space discussion on the paper IR3DE: A LINEAR ROUTER FOR LARGE LANGUAGE MODELS

Training-Free Composition of Expert Models [↗](#)

May 2026

X article from Gensyn — Coverage of invited X Space discussion on the paper TRAINING-FREE DYNAMIC UPCYCLING OF EXPERT LANGUAGE MODELS

Languages ---

Italian - native

English - full professional proficiency

Spanish - elementary proficiency

Coding skills ---

Python, C, PyTorch, Numpy, Scikit-Learn, WandB, Matplotlib, HTML, HTML5, HTML Bootstrap, CSS, Jax, FedJax, Haiku, Flax, Flower, Textual, Java, C++, Bash, Matlab, Javascript, ReactJS, Vite, PHP, R, Apache Spark, Apache Hadoop